

Mrityunjay Kumar

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Core ML engineer focused on the reward and evaluation side of RL for tool-using agents: reward signal correctness, LLM-as-judge grading, multi-turn evaluation, and multimodal tool integration into live RL rollouts. Led early-stage development and evaluation of **Rolis** (EuroSys 2022). Patent holder in storage metadata verification.

SKILLS

- **Core ML:** RL reward & evaluation infrastructure, RLVR/RLHF/SFT, LLM-as-judge reward design, multi-turn agentic evaluation, multimodal grading, RL environment & tool integration, async rollout integration, LLM fine-tuning pipelines, prompt optimization
- **Distributed Systems & Infra:** Distributed systems, consensus protocols (Paxos, Raft), transactional databases, container orchestration, cloud-native microservices, Redis, high-throughput pipeline design
- **Cloud & DevOps:** Azure (ACI, Blob, ML, AAD), AWS, Docker, Prometheus, Grafana, OpenTelemetry/Jaeger, Nginx, gRPC, REST
- **Languages:** Python, C#, C++, C, Java · PyTorch, Git

EXPERIENCE

Microsoft

05/2022 – Present

Senior ML Engineer, RL Reward & Evaluation Infrastructure · PowerPoint document agent

Mountain View, CA

◦ RL Reward Signal & Evaluation Methodology

- * Diagnosed a corrupt reward signal in the document agent's RL training loop: single-turn evaluation was structurally blind to turn-level error propagation, tool-call convergence failure, and stale-context degradation. Designed a multi-turn *simulation* methodology to fix it, replaying real agent trajectories against live tool execution in a controlled environment to validate reward-signal quality before it enters training.
- * Built *LLM-as-judge* reward grading for multi-turn agent trajectories.
- * Built a document-render-to-image visual grading pipeline; **Etag/IfNotModified** optimistic concurrency across parallel graders eliminates duplicate rewards at trajectory scale.
- * Drove an org-wide checkpoint pre-screening gate: RL candidates simulated before end-to-end integration, eliminating wasted compute on non-viable checkpoints; simulation-grounded comparisons directly informed which checkpoints advanced to full RL training.
- * Established the first quantified per-slide latency profile (33 to 75s/slide) by isolating LLM reasoning (7–30s) from tool execution (5–8s/turn) under a 5-turn cap, enabling model-versus-tool regression attribution invisible to end-to-end metrics.

◦ Multimodal RL Environment & Tool Execution

- * Integrated an image-generation engine as an agent-callable tool in the RL rollout, extending the document agent's action space to a new modality.
- * Owned the full tool-execution stack for RL training on the document agent: client-to-daemon architecture, per-trajectory VM provisioning, and an asynchronous multimodal pipeline. Scaled to 7,000 containers while holding SLOs (P50 under 60s, P95 under 120s, P99 under 180s), unblocking concurrent RL pipelines across teams.
- * Unified Anthropic Claude, Azure OpenAI (incl. O-series reasoning models), and internal production serving endpoints behind a common provider-adapter interface for the rollout/eval loop.

◦ Production Platform, Scale & Observability

- * Engineered the platform running the simulation at scale: nginx-balanced Python orchestration tier over a .NET/COM PowerPoint execution tier, Redis metadata store + distributed queue, artifact storage, and a cluster management UI/API; deployed as a 20+ container system on ACI with per-service health checks.
- * Achieved **5.7x throughput** (0.70 to 4.01 files/s) and **7.8x worker utilization** (12% to 94%) by resolving a three-stage bottleneck chain: base64 upload elimination (+73%), COM-pool redistribution (util 17% to 72%), and CPU quotas to prevent cache thrashing (+24%). 200-file batch 285s to 49.8s. Fleet projection: 100 VMs at 301 files/s, approx. 1M files/hr.
- * Resolved two RL-scale blockers (platform throttling at 1.4M+ requests/10 min; VM-lifecycle resource leak blocking 20k-VM training+grading parallelism); instrumented end-to-end with OpenTelemetry/Jaeger tracing that makes multi-turn eval trajectories attributable per turn (queue wait, worker execution, and artifact emission), alongside Prometheus + Grafana.

- **ML Platform & Fine-Tuning Data Pipeline**

- * Authored a model-first platform architecture (ingestion, grading/tagging, eval, observability) adopted as org blueprint; adjacent teams directed to use rather than rebuild.
- * Designed a multi-turn SFT data strategy: three-phase curation (programmatic tagging to expert audit to automation), full training schema, DAG execution optimization cutting agent latency from approx. 100s to 37s.

VMware Inc.

07/2020 – 05/2022

Member of Technical Staff III

Palo Alto, CA

- **Distributed File System: Storage Layer**

- * Designed and implemented distributed FSCK for post-crash metadata repair: cross-consistency matrices across B+ trees, bitmaps, and segment usage tables.
- * Built cloud-native microservice for on-demand/scheduled integrity validation of the file system k-v store; extended to snapshot, unmap, and segment-cleaning checks.
- * Implemented aggregated snapshot capacity statistics collection from scratch; p99 latency 30ms.

Distributed Systems Lab, Stony Brook University

06/2019 – 05/2020

Graduate Research Assistant, [Prof. Shuai Mu](#)

EuroSys 2022

- **Rolis: Replicated Multi-core Transactional Database Engine**

- * Implemented core engine mechanisms (see Publications): async multi-process Paxos replication and multi-core log truncation for checkpointing. Profiled CPU/heap/memory via gperftools and mutrace; built a header-only C++ logging library with aligned allocators and auto-flush.

Talentica Software

04/2016 – 01/2019

Senior Software Engineer

Pune, India

- Built real-time ML inference pipeline (Apache Storm, Kafka, Cassandra, AWS) for BLE indoor location prediction; feedback loop improved accuracy by 95% and a live calibration portal improved site correctness by 70%.

MediaTek

08/2014 – 04/2016

Software Engineer

Noida, India

- Audio module (C/C++): reduced playlist overload by 25%, integrated BT stack into MMI layer, authored GPS-WiFi-BT combo stress-test tooling.

PUBLICATIONS & PATENTS

- **Rolis: a software approach to efficiently replicating multi-core transactions** [[EuroSys 2022](#)] (**acknowledged contributor**): Designed and implemented core mechanisms for a distributed multi-core transactional database engine: asynchronous Paxos replication with minimal throughput loss, multi-core log truncation for checkpointing enabling read-only transactions at 10M ops/sec, and a replay protocol guaranteeing serializability across in-memory and disk logs.
- **Verification of metadata consistency across snapshot COW B+ tree logical maps** [[US11573860B1](#)]: Methodology for verifying consistency of snapshot metadata across a hierarchy of ordered data structures; covers cross-tree node identification, consistency matrix construction, and repair logic.
- **Learning to Fingerprint the Latent Structure in Question Articulation** [[publication](#)]: Demonstrated that latent question patterns can be modeled as a cost-function-maximizing system approximated by a trained neural auto-encoder.

EDUCATION

- **Stony Brook University**: M.S. in Computer Science (2019–2020)
- **NIT Bhopal**: B.Tech. in Computer Science & Engineering (2010–2014)